

STEERING

REFER TO 1990 CELICA REPAIR MANUAL (Pub. No. RM149U)

NOTE: The following pages contain only the points which differ from the above listed manual.

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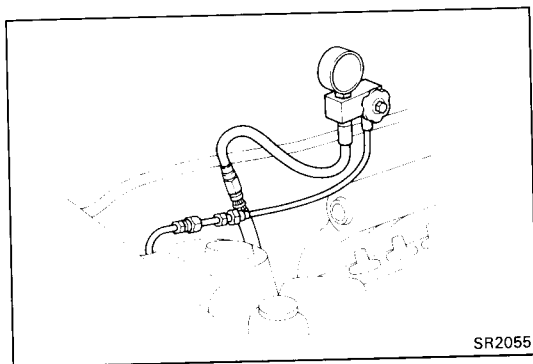
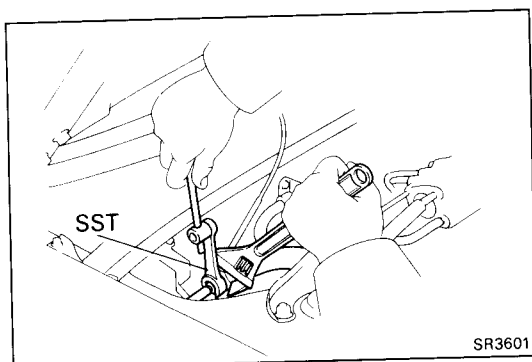
POWER STEERING

On-Vehicle Inspection

OIL PRESSURE CHECK

1. CONNECT PRESSURE GAUGE

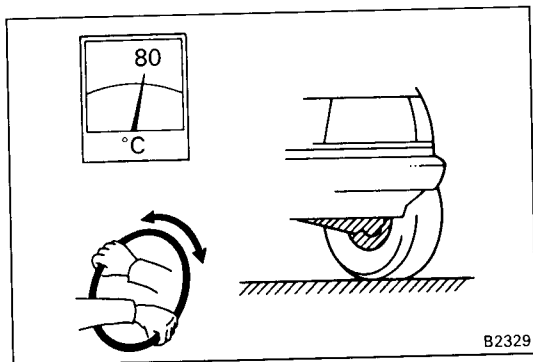
- (a) Using SST, disconnect the pressure line joint.
SST 09631-22020



- (b) Connect the gauge side of the pressure gauge to the PS pump side, and the valve side to the gear housing side.
- (c) Bleed the system. Start the engine and turn the steering wheel from lock to lock two or three times.
- (d) Check that the fluid level is correct.

2. CHECK THAT FLUID TEMPERATURE IS AT LEAST 80°C (176°F)

3. START ENGINE AND RUN IT AT IDLE



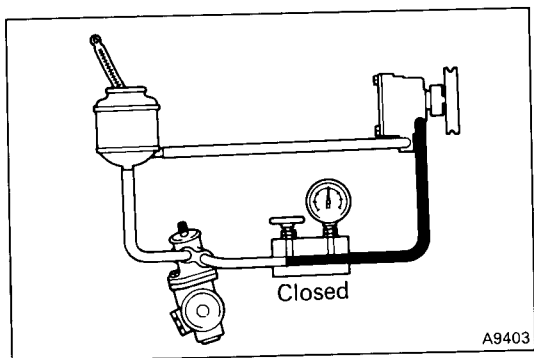
4. CHECK FLUID PRESSURE READING WITH VALVE CLOSED

Close the pressure gauge valve and observe the reading on the gauge.

Minimum pressure: 80 kg/cm² (1,138 psi, 7,845 kPa)

NOTICE:

- Do not keep the valve closed for more than 10 seconds.
 - Do not let the fluid temperature become too high.
- If pressure is low, repair or replace the PS pump.



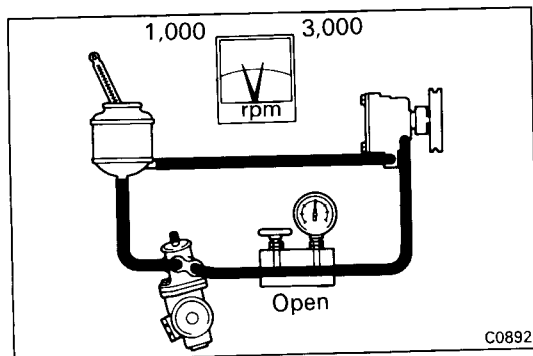
5. OPEN VALVE FULLY

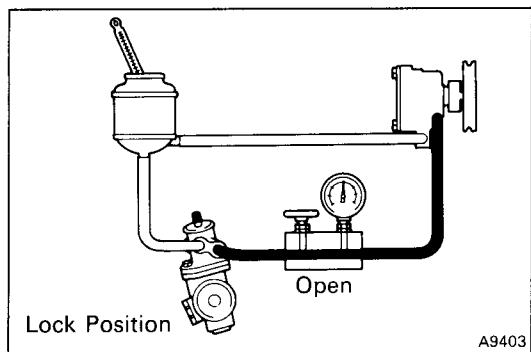
6. CHECK AND RECORD PRESSURE READING AT 1,000 RPM

7. CHECK AND RECORD PRESSURE READING AT 3,000 RPM

Check that there is 5 kg/cm² (71 psi, 490 kPa) or less difference in pressure between the 1,000 rpm and 3,000 rpm checks.

If the difference is excessive, repair or replace the flow control valve of the PS pump.





8. CHECK PRESSURE READING WITH STEERING WHEEL TURNED TO FULL LOCK

Be sure the pressure gauge valve is fully opened and the engine idling.

Minimum pressure: 80 kg/cm² (1,138 psi, 7,845 kPa)

NOTICE:

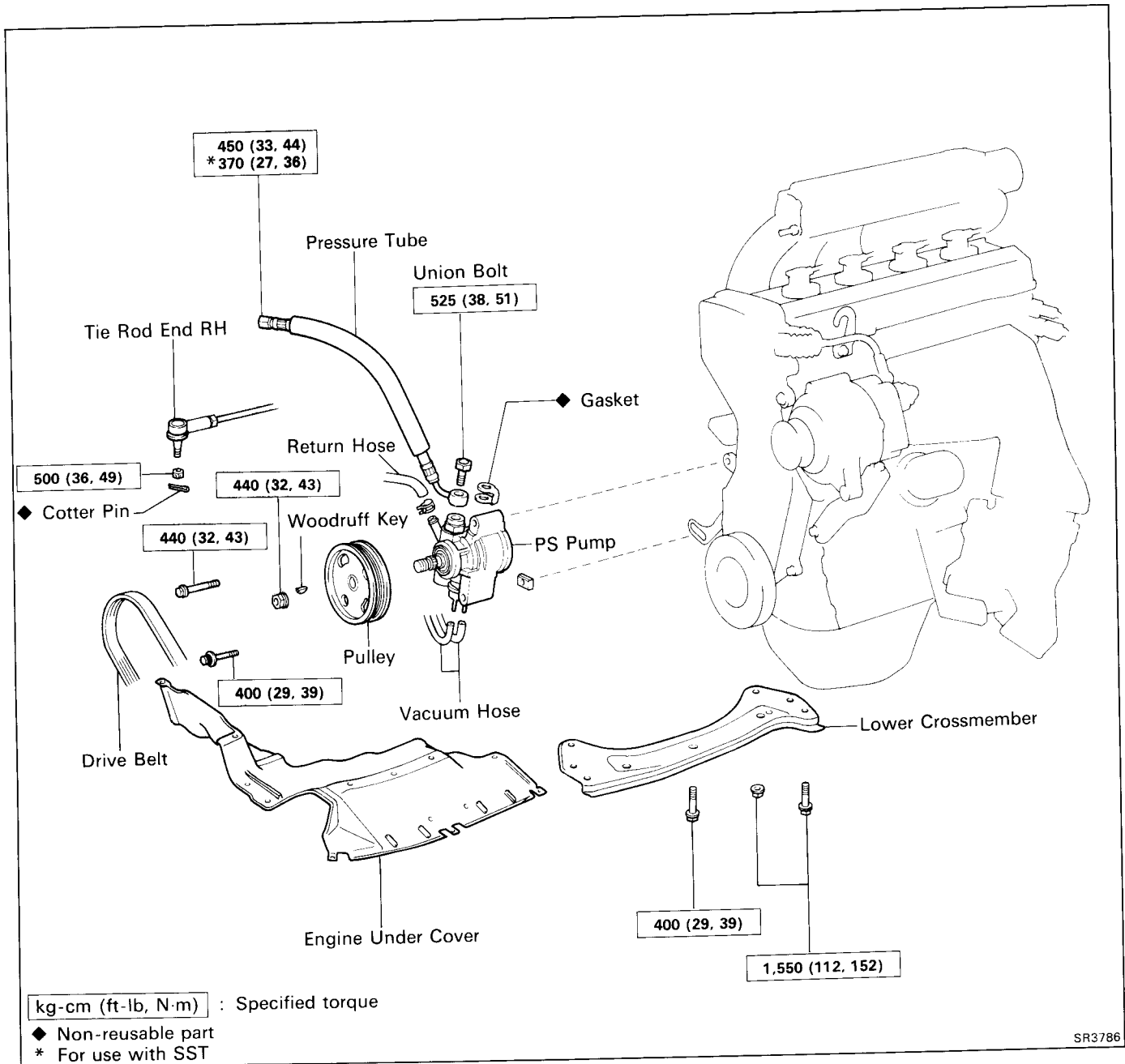
- Do not maintain lock position for more than 10 seconds.
- Do not let the fluid temperature become too high.

If pressure is low, the gear housing has an internal leak and must be repaired or replaced.

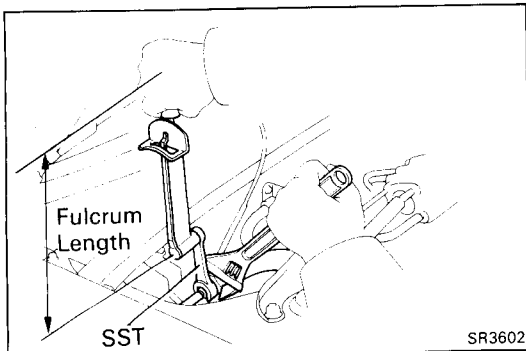
Power Steering Pump

REMOVAL AND INSTALLATION OF POWER STEERING PUMP

Remove and install the parts as shown.



SR3786



SR3602

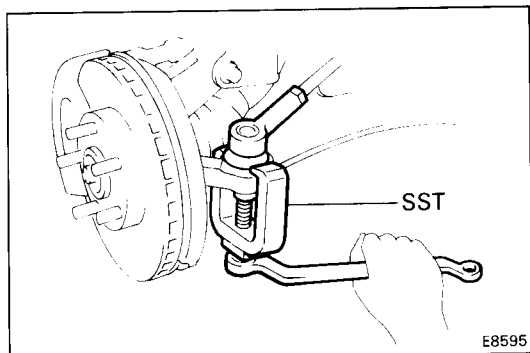
(MAIN POINTS OF REMOVAL AND INSTALLATION)

1. DISCONNECT AND CONNECT PRESSURE LINE

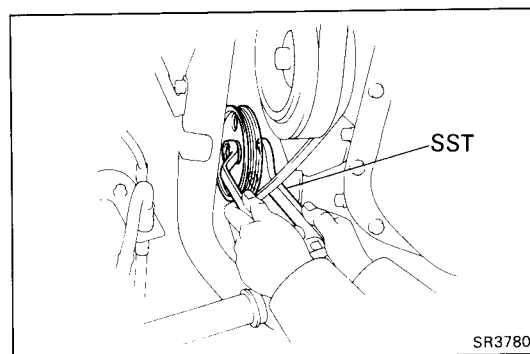
Using SST, disconnect and connect the pressure line.
SST 09631-22020

Torque: 370 kg-cm (27 ft-lb, 36 N·m)

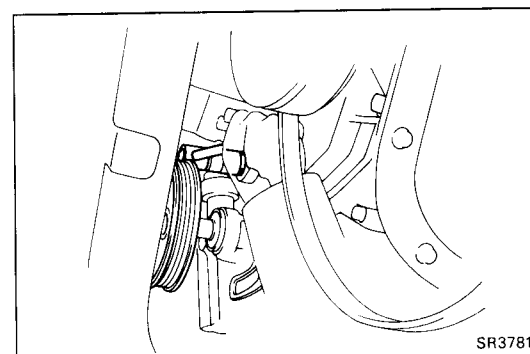
HINT: When connecting, use a torque wrench with a fulcrum length of 300 mm (11.81 in.).

**2. DISCONNECT TIE ROD END RH**

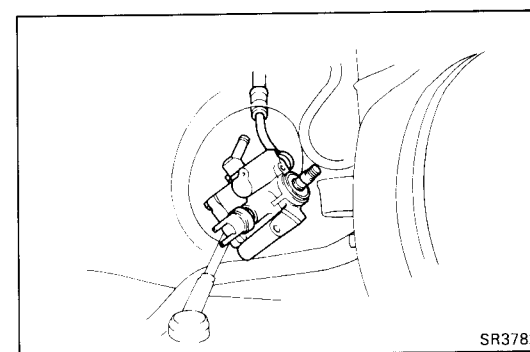
- (a) Remove the cotter pin and nut.
 - (b) Using SST, disconnect the tie rod end RH.
- SST 09611-22012

**3. REMOVE PS PUMP WITH PRESSURE TUBE**

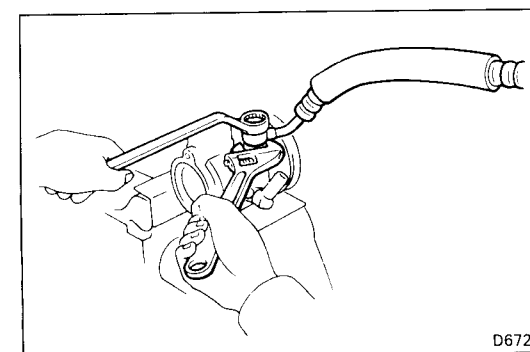
- (a) Using SST to hold the pulley, loosen the pulley set nut.
- SST 09616-30020



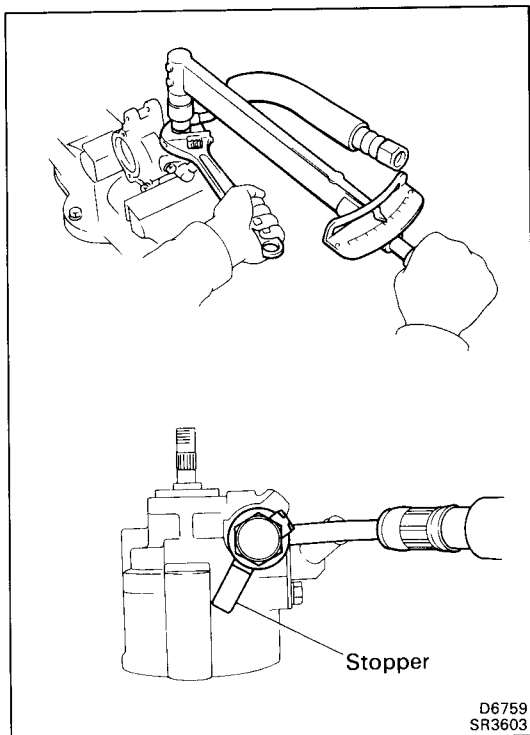
- (b) Slide the pump pulley, and loosen the through bolt.



- (c) Remove the PS pump from the tie rod end hole.

**4. REMOVE PRESSURE TUBE**

- (a) Mount the PS pump in the vise.
- NOTICE: Do not tighten the vise too tight.**
- (b) Remove the union bolt and pressure tube.

**5. INSTALL PRESSURE TUBE**

- (a) Install the pressure tube with the union bolt and a new gasket.
- (b) Make sure the stopper is touching the pump housing as shown, then torque the union bolt.

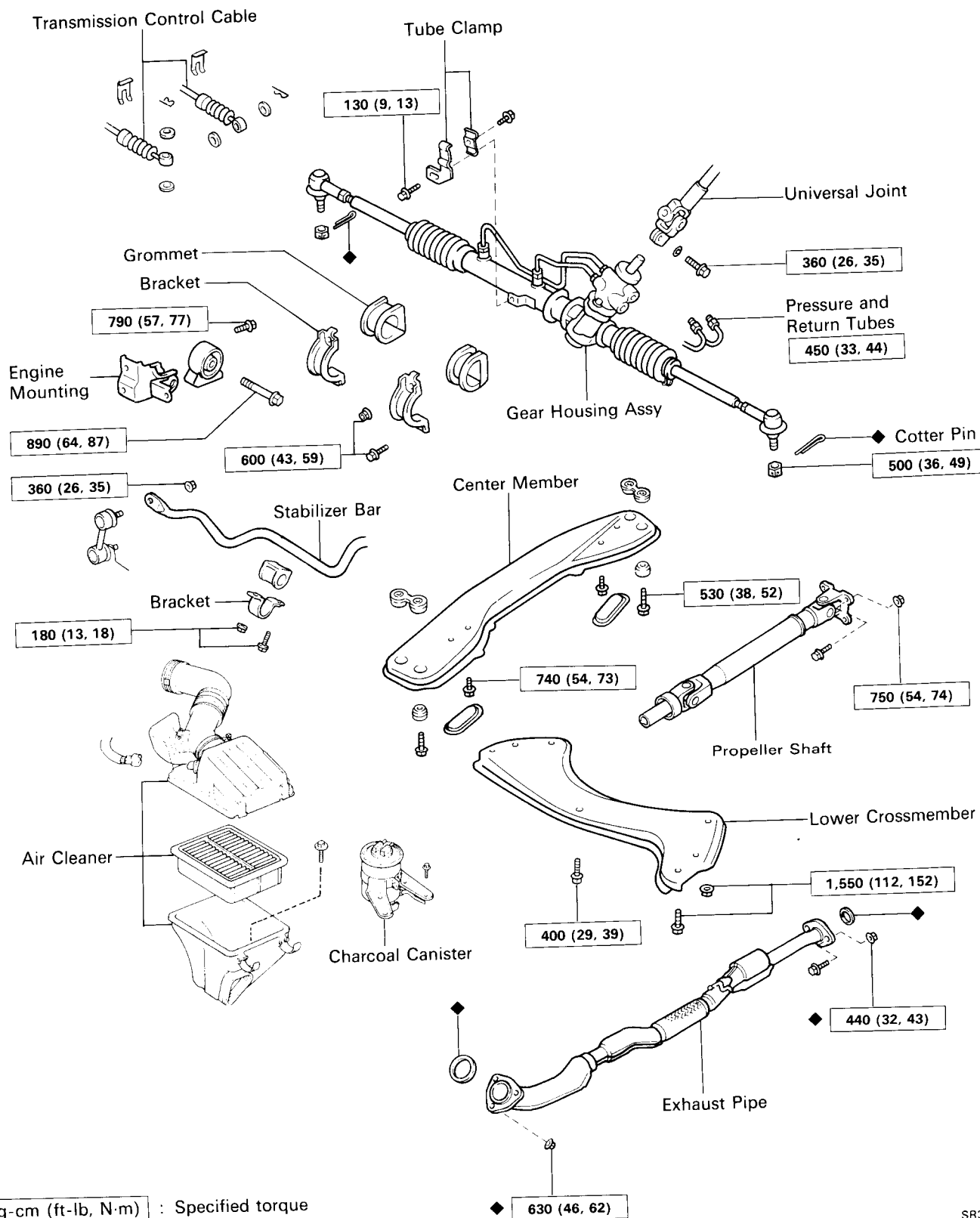
Torque: 525 kg-cm (38 ft-lb, 51 N·m)

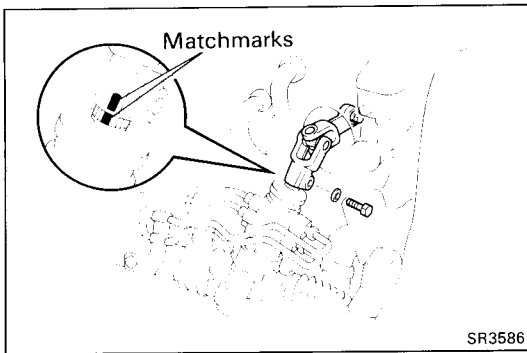
6. ADJUST DRIVE BELT TENSION AFTER INSTALLING PS PUMP**7. BLEED POWER STEERING SYSTEM**

Gear Housing

REMOVAL AND INSTALLATION OF STEERING GEAR HOUSING

Remove and install the parts as shown.

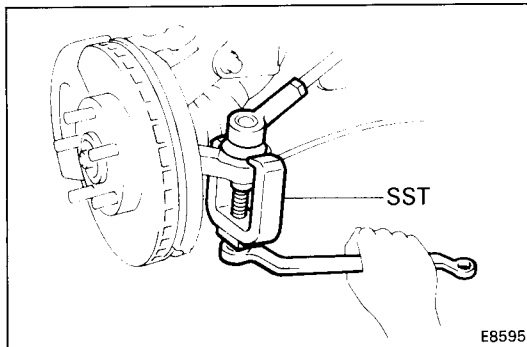




(MAIN POINTS OF REMOVAL AND INSTALLATION)

1. DISCONNECT UNIVERSAL JOINT

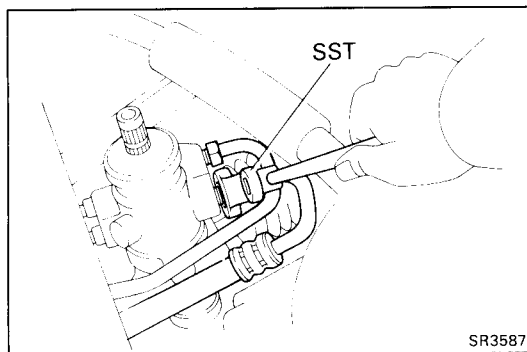
- (a) Place matchmarks on the universal joint and control valve shaft.
- (b) Loosen the bolt on the upper side of the universal joint, remove the bolt on the lower side and disconnect the universal joint.



2. DISCONNECT TIE ROD ENDS

- (a) Remove the cotter pin and nut.
- (b) Using SST, disconnect the tie rod end from the knuckle arm.

SST 09611-22012

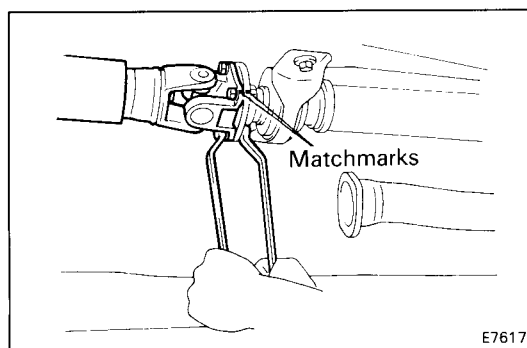


3. DISCONNECT AND CONNECT PRESSURE AND RETURN LINE

Using SST, disconnect and connect the pressure and return line.

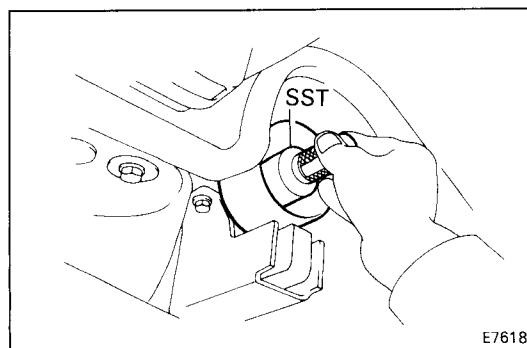
SST 09631-22020

Torque: 450 kg-cm (33 ft-lb, 44 N·m)



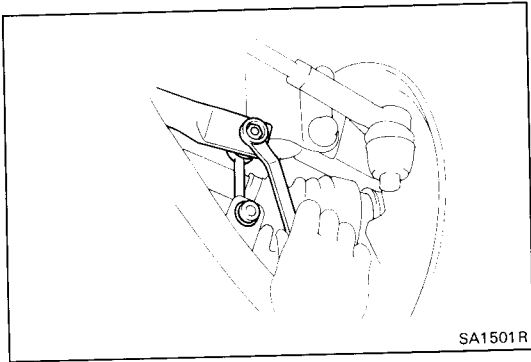
4. REMOVE PROPELLER SHAFT

- (a) Place matchmarks to the propeller shaft flange and intermediate shaft flange.
- (b) Remove the four bolts and nuts.

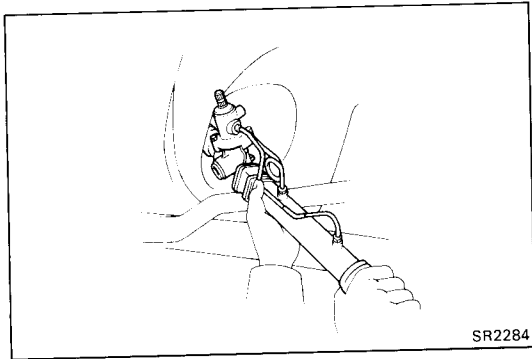


- (c) Pull out the propeller shaft and insert SST.

SST 09325-20010

**5. DISCONNECT STABILIZER BAR RH**

- (a) Remove the stabilizer bar bracket RH.
- (b) Disconnect the stabilizer bar from the link.

**6. REMOVE GEAR HOUSING**

- (a) Remove the two bolts, two nuts and the two brackets.
- (b) Remove the PS tube clamp bolt.
- (c) Move the gear housing to the right side.
- (d) Remove the gear housing through the RH hole.

7. CHECK STEERING WHEEL CENTER POINT**8. CHECK TOE-IN**